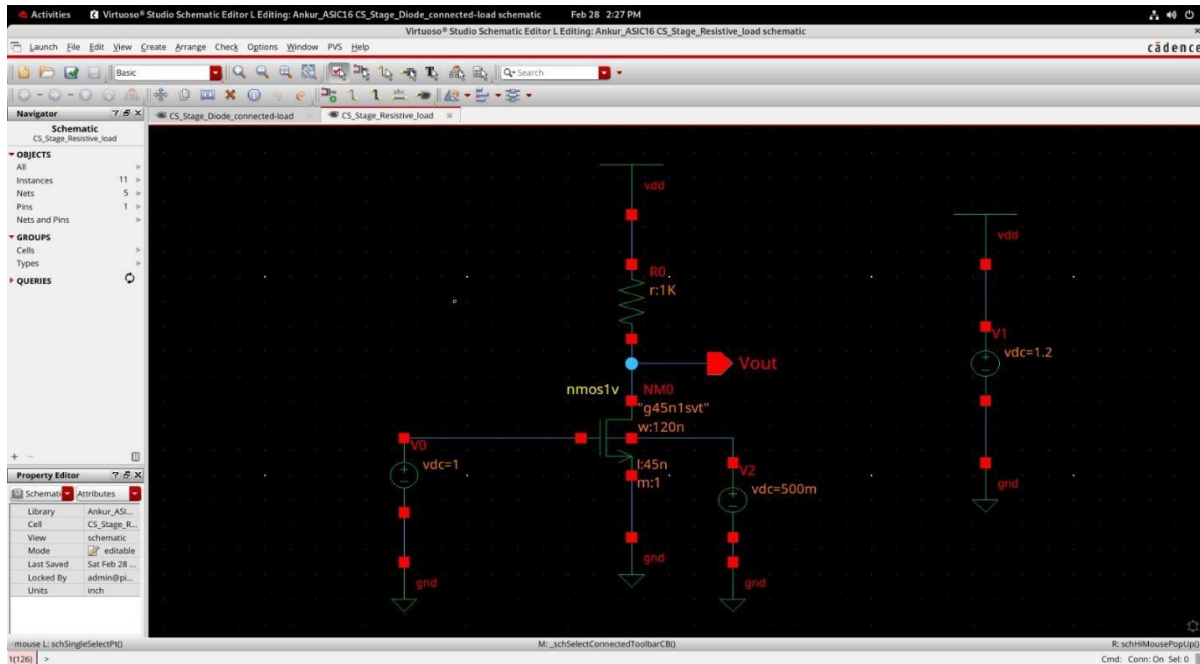


Body Effect Analysis

Analysing the effect of Body Voltage on CS stage amplifier with Resistive Load



Observation Table:

Vb	Vin	Vth	Sat region	Id	Vout
0V	822.24mV	586.64mV	600mV-1V	24.97uA	1.17V
-0.5V	822.57mV	640.85V	660mV-1V	17.76uA	1.18V
+0.5V	822.57mV	481.88mv	500mV-1V	40.74uA	1.15V

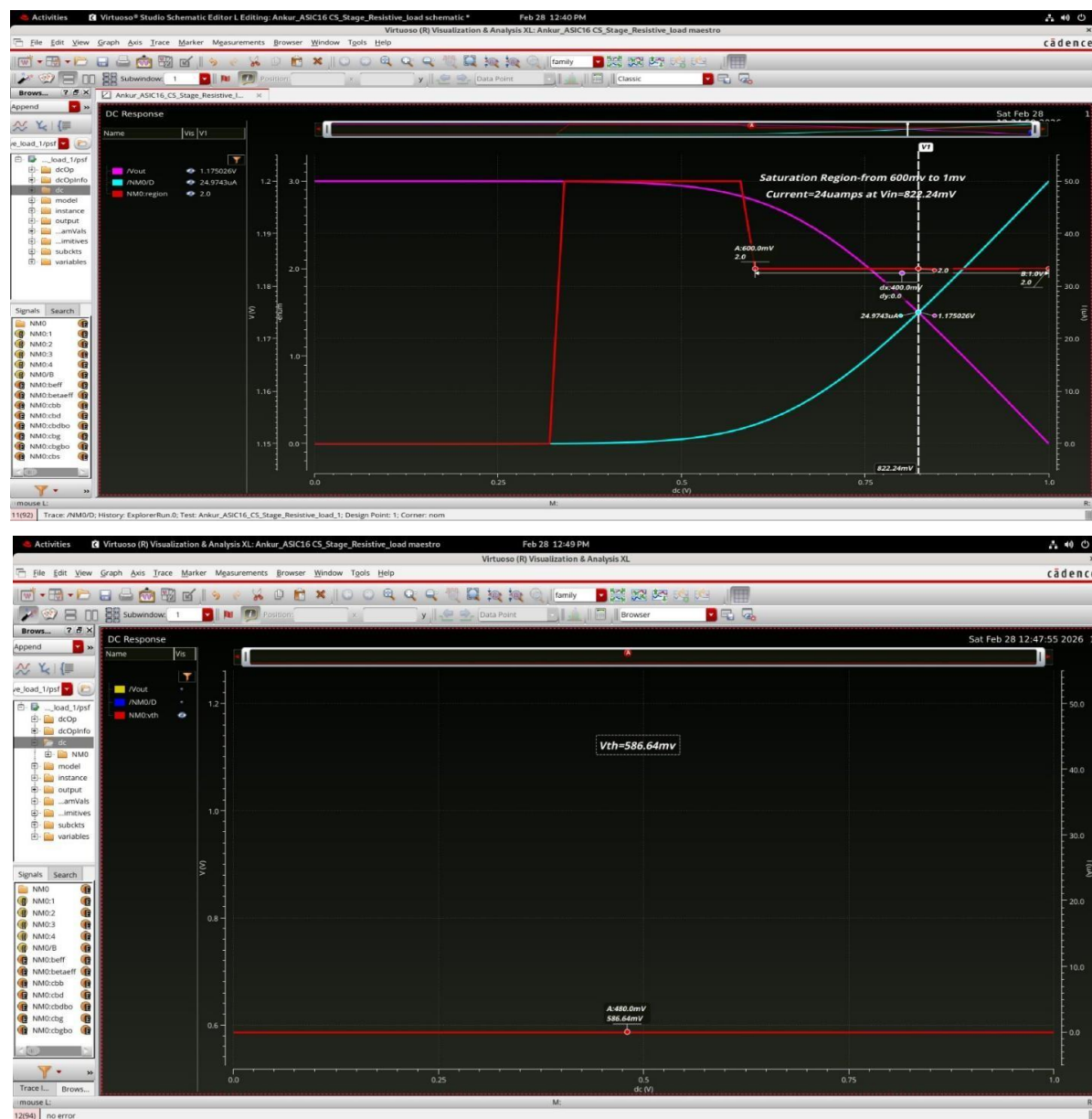
Conclusion: Case 1: For $V_b = -0.5V$ $V_{sb} = V_s - V_b = 0 - (-0.5) = +0.5V (+ve)$

- R.B. will increase
- V_{th} will increase
- I_d will decrease
- V_{out} will increase
- V_{dsat} will increase
- Sat Region will shift rightwards

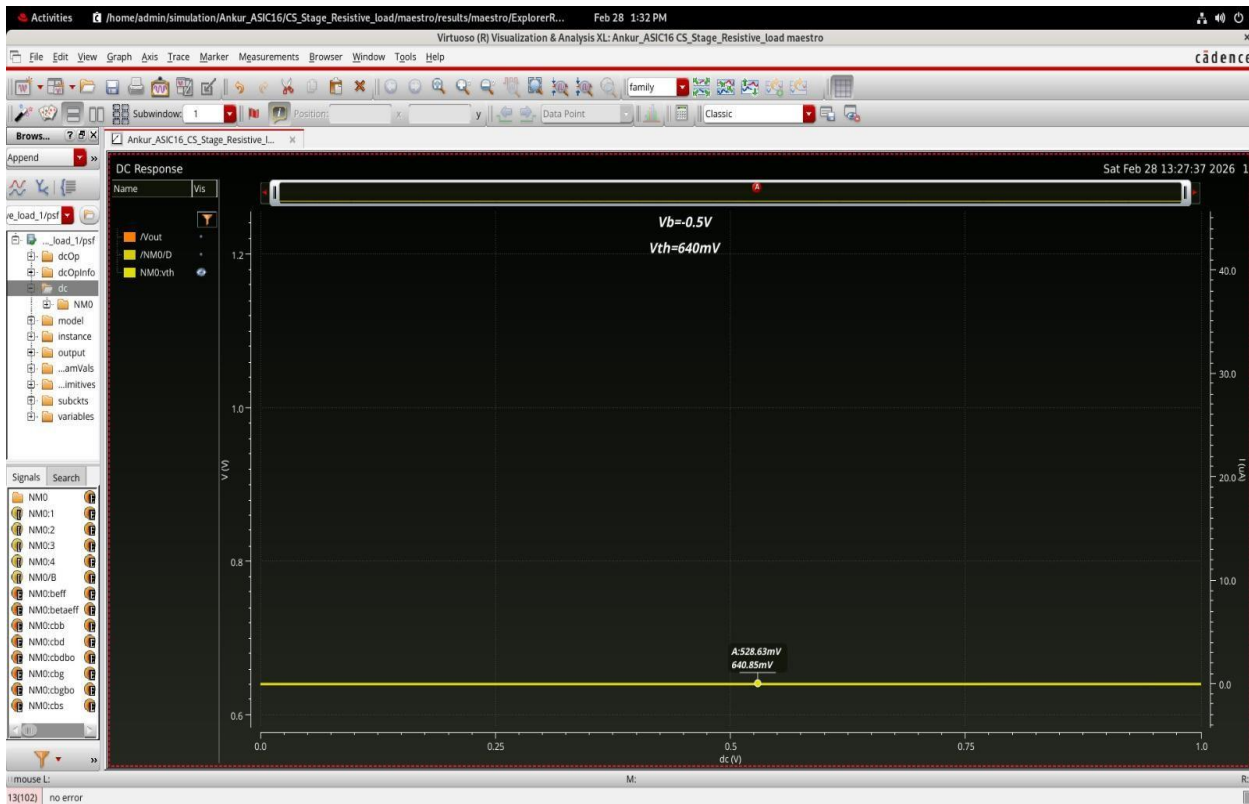
Case 2 : For $V_b = +0.5V$; $V_{sb} = V_s - V_b = 0 - (+0.5V) = -0.5V(-ve)$

- R.B. will decrease
- V_{th} will decrease
- I_d will increase
- V_{out} will decrease
- V_{dsat} will decrease
- Sat Region will shift leftwards

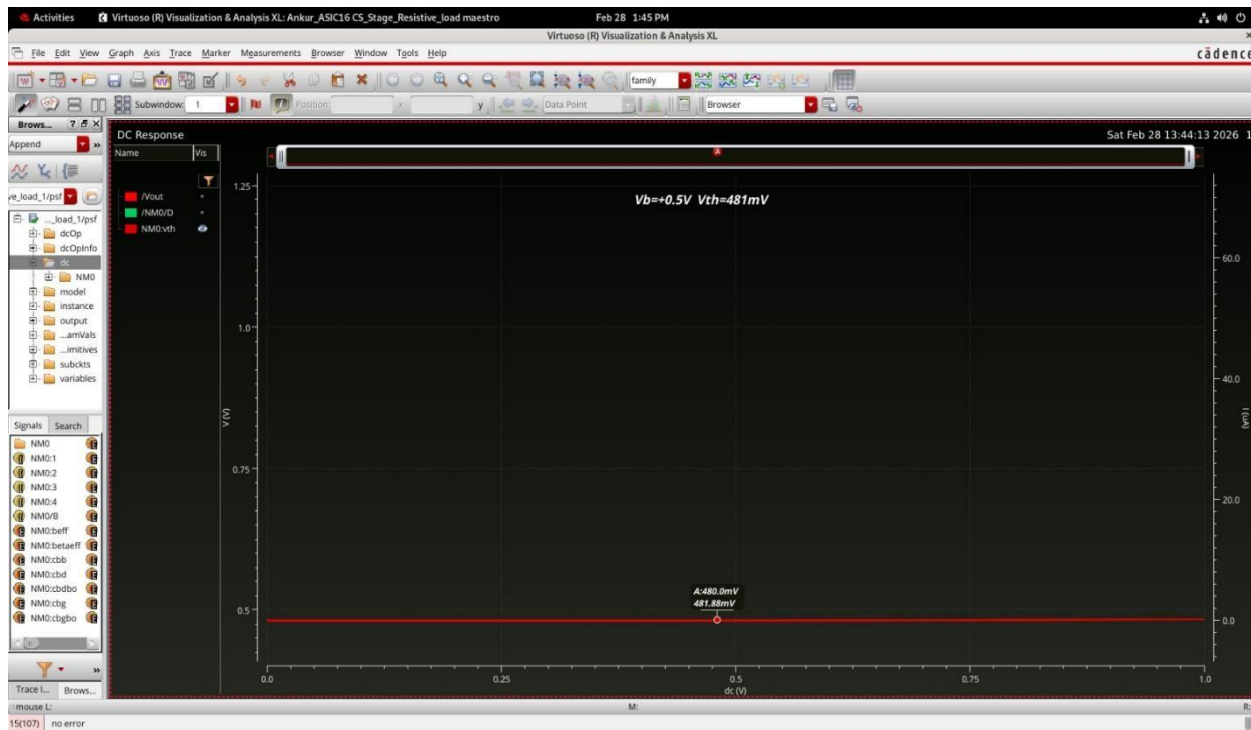
When $V_b = 0V$



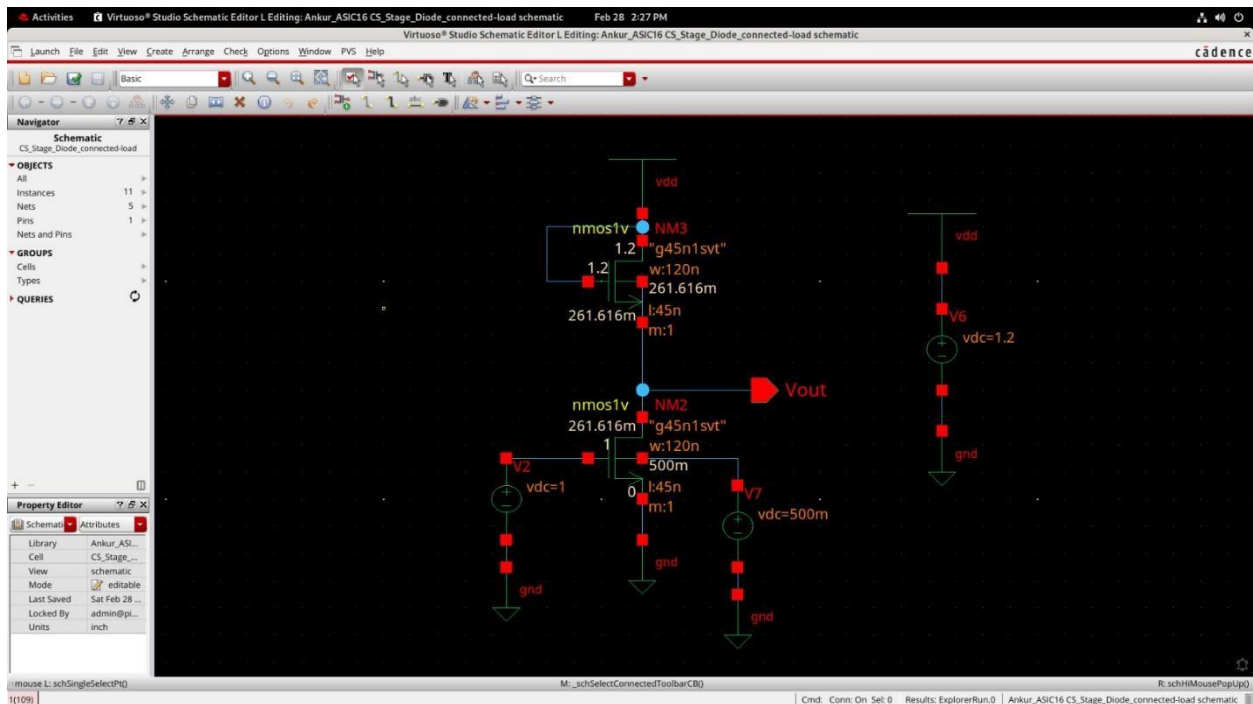
When $V_b = -0.5V$



When $V_b = +0.5V$



Analysing the effect of Body Voltage on CS Stage amp with diode connected load:



Observation Table:

Vb	Vin	Vth	Sat region	Id	Vout
0V	704.32mV	591.22mV	600mV - 1V	8.22uA	508.26mV
-0.5V	704.04mV	641.81mV	660mV - 1V	4.78uA	557.61mV
+0.5V	704.15mV	487.15mV - 521.86mV	520mV - 920mV	15.17uA	437.29mV

Conclusion:

Case 1: Body voltage decreased ($V_B = -0.5V$)

Since source is at 0 V:

$$V_{SB} = V_S - V_B$$

$$V_{SB} = 0 - (-0.5) = +0.5V$$

Reverse body bias increases

Body effect increases V_{TH} Threshold equation:

$$V_{TH} = V_{TH}^0 + \gamma (\sqrt{2\phi + V_{SB}} - \sqrt{2\phi})$$

Que. Why current decreases?

$$I_D \propto (V_{GS} - V_{TH})^2$$

$$\text{If } V_{TH} \uparrow (V_{GS} - V_{TH}) \downarrow$$

So, $I_D \downarrow$

Que. Why V_{out} increases?

Diode-connected load current:

$$I_D = \frac{1}{2} k (V_{OUT} - V_{TH(load)})^2$$

Now driver current decreased.

To maintain current equality:

$$I_{driver} = I_{load}$$

Case 2: Body voltage increased ($V_B = +0.5V$)

$$V_{SB} = 0 - 0.5 = -0.5V$$

Body bias reduced Threshold voltage decreases Current increases

Saturation region shifts left

$$V_{TH} \quad V_{OUT} \quad V_{TH} \downarrow$$

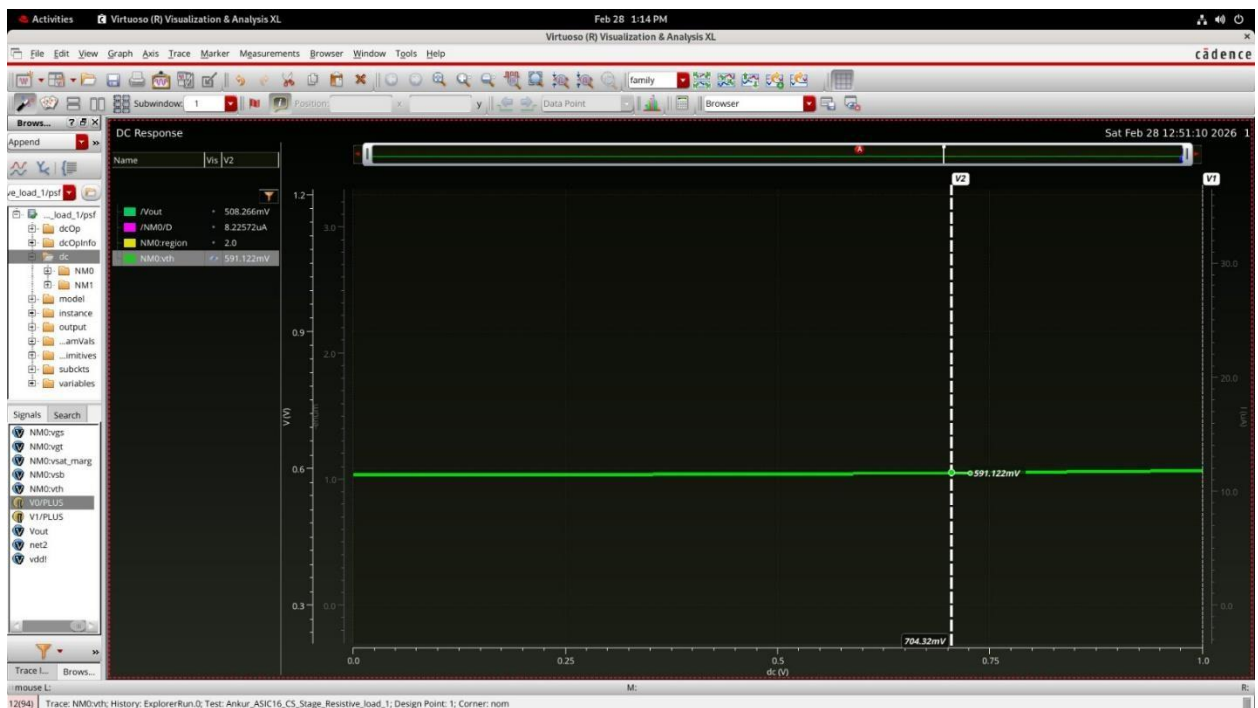
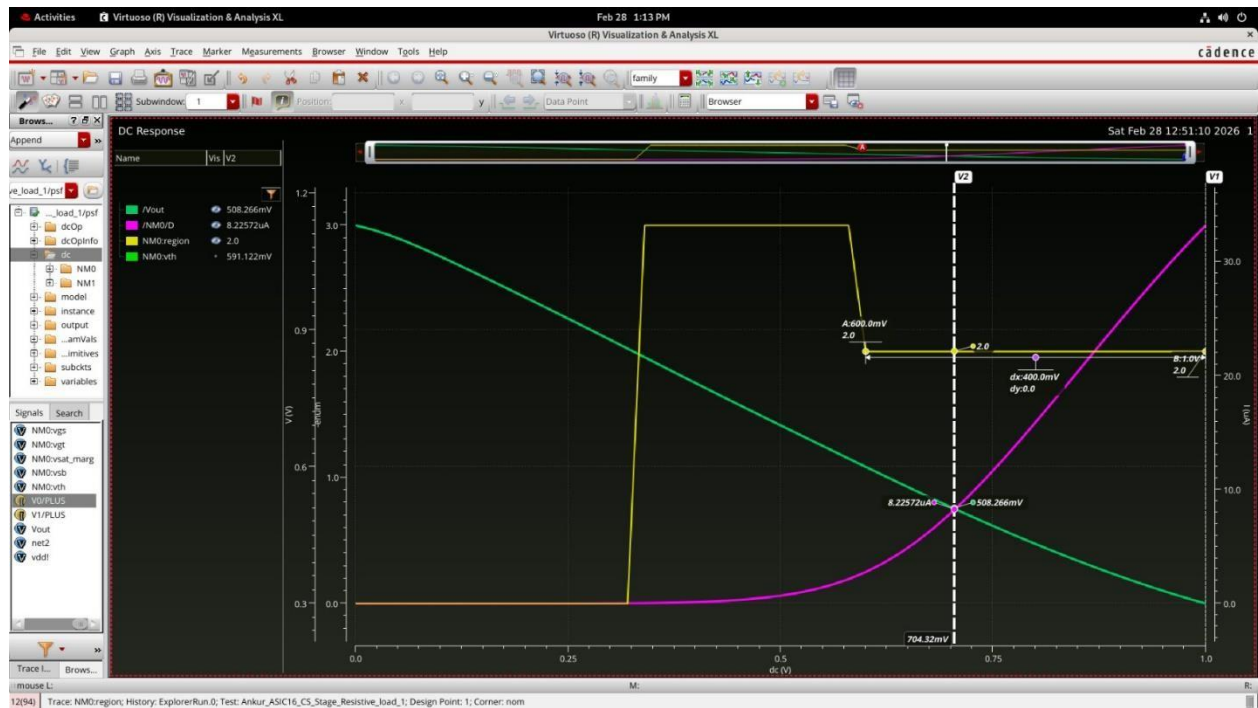
$$(V_{GS} - V_{TH}) \uparrow \text{ So: } I_D \uparrow$$

To balance with diode load: Load must carry higher current.

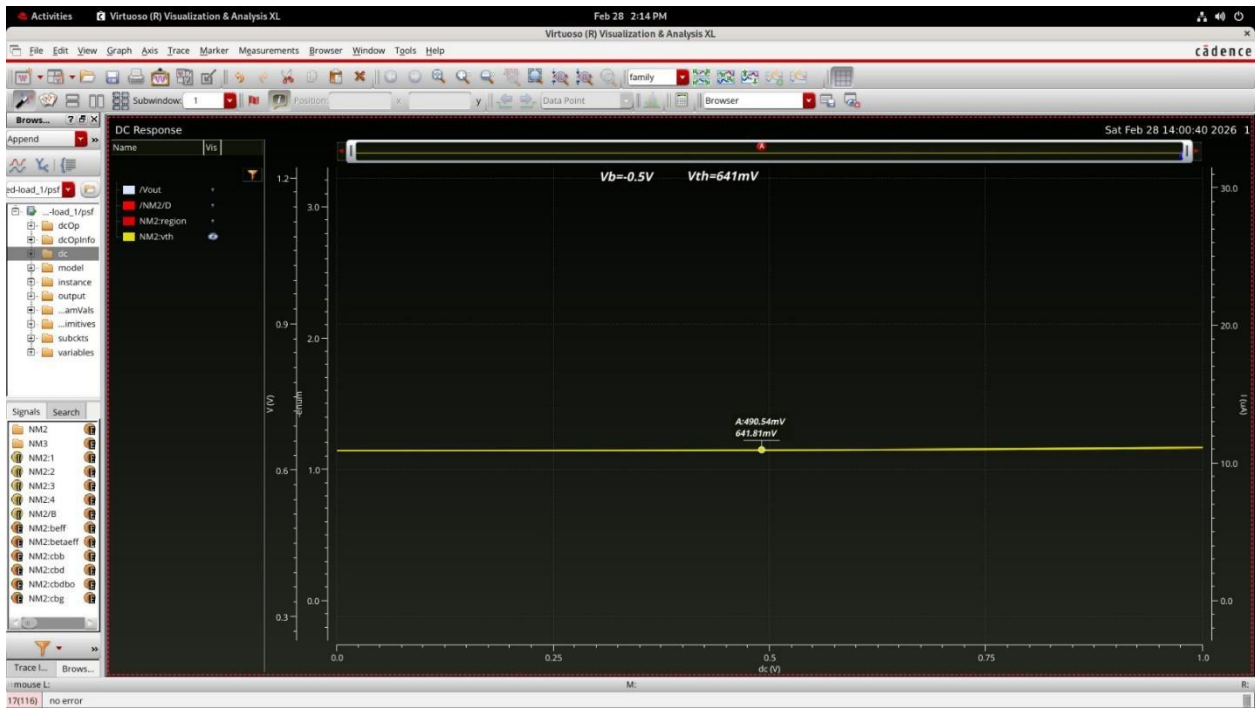
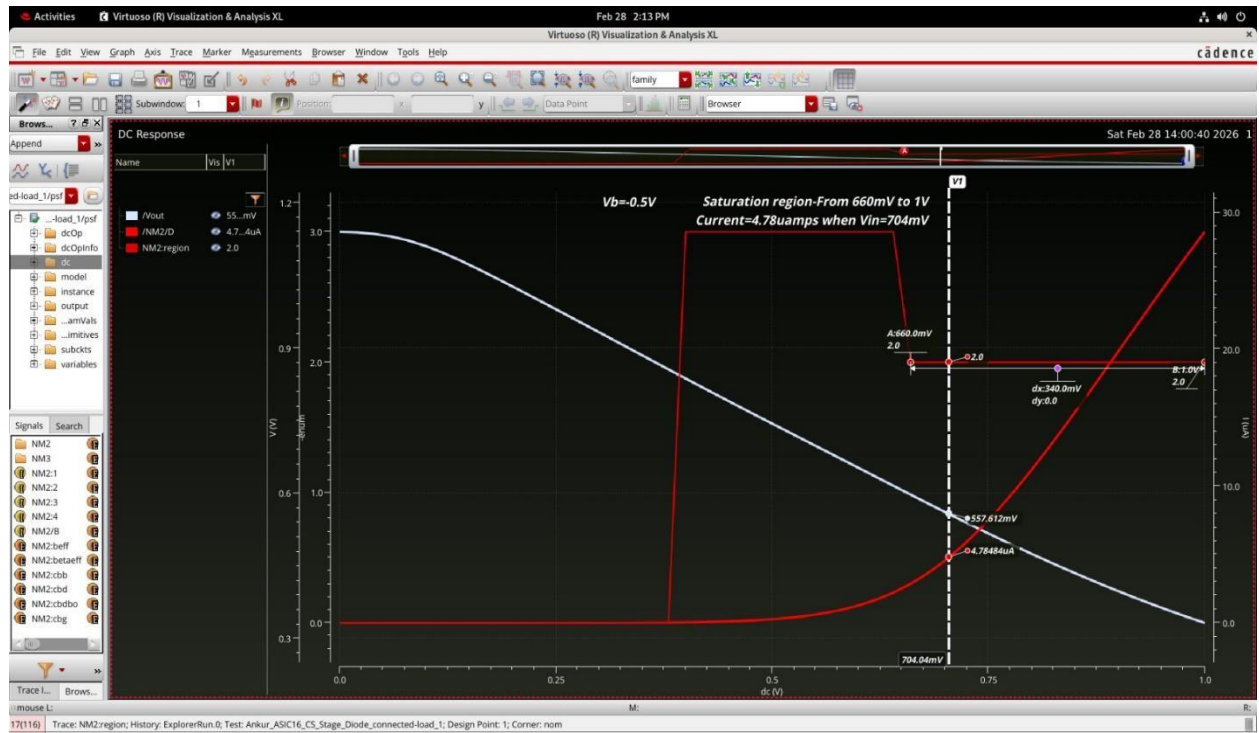
Higher current in diode-connected NMOS requires:

$$V_{OUT} \downarrow$$

When $V_b=0V$



When $V_b = -0.5V$



When $V_b = +0.5V$

